Combined Oxidative Phosphorylation Deficiency 52 (COXPD52): A Comprehensive Disease Characteristics Report

Disease: Combined Oxidative Phosphorylation Deficiency 52 (COXPD52) **Causal gene: NFS1** (cysteine desulfurase, 20q11.22; UniProt Q9Y697; NCBI Gene 9054; HGNC:7808) **MONDO ID:** MONDO:0030311 · **OMIM:** #619386 · **Synonym:** COXPD52 **Inheritance:** Autosomal recessive · **Category:** Mendelian, ultra-rare mitochondrial disease

Summary

Combined Oxidative Phosphorylation Deficiency 52 (COXPD52) is an ultra-rare autosomal recessive mitochondrial disease caused by biallelic hypomorphic variants in *NFS1*, the nuclear gene encoding the sole mitochondrial **cysteine desulfurase**—the enzyme that initiates iron–sulfur (Fe–S) cluster biogenesis. Because NFS1 mobilizes sulfur from L-cysteine to build the [2Fe–2S] clusters that mature downstream Fe–S enzymes, reduction of its activity produces a *combined* deficiency affecting multiple mitochondrial and extra-mitochondrial Fe–S-dependent processes simultaneously. The disease was first defined in 2014 under the descriptive name "infantile mitochondrial complex II/III deficiency," and later subsumed under the OMIM COXPD series as entry #619386.

The clinical spectrum is **genotype-correlated and strikingly variable**. At the severe end, infants homozygous for the recurrent missense variant **c.215G>A (p.Arg72Gln)** present in the neonatal/infantile period with lactic acidemia, hypotonia, muscle weakness, abnormal mitochondrial morphology on muscle biopsy, combined respiratory-chain complex II+III (variably I) deficiency, and multisystem organ failure that is frequently fatal. At the mild end, a compound heterozygous patient (p.Arg72Gln / p.Arg412His) survived to present as a young adult with gastrointestinal symptoms and fatigue. This phenotypic breadth reflects NFS1's position as the **rate-limiting, upstream node** of the entire cellular Fe–S supply chain: its sulfur

output feeds the respiratory chain, the TCA-cycle aconitases, lipoic acid synthesis, the cytosolic iron–sulfur assembly (CIA) pathway (including DNA replication/repair enzymes), and tRNA thiolation.

There is **no disease-specific or approved therapy**; management is supportive mitochondrial care (treatment of lactic acidosis, nutritional support, cofactor/"mitochondrial cocktail" supplementation, and avoidance of metabolic decompensation). Diagnosis rests on **exome or genome sequencing** demonstrating biallelic *NFS1* variants, supported by a characteristic biochemical signature: elevated blood/CSF lactate, abnormal mitochondria on muscle biopsy, combined complex II/III (±I) deficiency, and—distinctively—reduced activity of **both mitochondrial and cytosolic aconitases**, which separates COXPD52 from mtDNA-encoded respiratory defects. This report synthesizes eight confirmed findings across 35 reviewed papers, organized against the 15-section disease-characteristics template.

Key Findings

Finding 1 — COXPD52 is an autosomal recessive mitochondrial disease caused by biallelic *NFS1* variants

COXPD52 (OMIM #619386; MONDO:0030311) is caused by biallelic pathogenic variants in **NFS1** (cysteine desulfurase, chromosome 20q11.22). The founding report by Farhan et al. (2014) used autozygosity mapping combined with exome sequencing in a consanguineous family to identify the homozygous variant **c.215G>A, p.Arg72Gln** in an infant with what the authors termed "infantile mitochondrial complex II/III deficiency, a novel autosomal recessive mitochondrial disease characterized by lactic acidemia, hypotonia, respiratory chain complex II and III deficiency, multisystem organ failure and abnormal mitochondria" [PMID: 24498631]. The same study reported that "we identified c.215G>A, p.Arg72Gln in NFS1 as the likely causative mutation."

The disease entity and its causal gene were corroborated and expanded by Yang et al. (2022), who reported three additional unrelated patients: "The c.215G>A p.(Arg72Gln) variant in NFS1 has been previously reported to cause infantile mitochondrial complex II and III deficiency. We describe three additional unrelated patients with the same missense variant" [PMID: 35026043]. The recurrence of the *same* missense allele across unrelated families, together with autosomal recessive segregation, firmly establishes *NFS1* as the single causal gene for this Mendelian disorder.

> **Evidence type:** Human clinical / genetic. **Key PMIDs:** 24498631, 35026043.

Finding 2 — NFS1 is the catalytic core of the mitochondrial Fe–S assembly complex; p.Arg72Gln impairs cysteine desulfurase activity

NFS1 is the catalytic cysteine desulfurase at the heart of the mitochondrial **iron–sulfur cluster (ISC) assembly complex**. It functions in a multi-subunit core together with ISD11 (LYRM4), mitochondrial acyl carrier protein (ACP), and the scaffold protein ISCU2, and is allosterically activated by frataxin (FXN), with electrons supplied by ferredoxin-2 (FDX2). Maio et al. (2020) describe how structural work on "the human cysteine desulfurase NFS1, in complex with the LYR protein ISD11, the acyl carrier protein ACP, and the main scaffold ISCU, have shed light on the molecular interactions that govern initial cluster assembly on ISCU" [PMID: 31918395]. NFS1 abstracts sulfur from L-cysteine to build [2Fe–2S] clusters on ISCU2.

In patients, the functional consequences of reduced NFS1 activity are broad. Yang et al. (2022) found that "functional analyses demonstrated decreased activity in respiratory chain complex II and variably in complexes I and III. We found decreased mitochondrial and cytosolic aconitase activities but only mildly affected lipoylation" of PDH/2-OGDH [PMID: 35026043]. The p.Arg72Gln residue lies adjacent to the catalytic machinery; its substitution reduces—but does not abolish—desulfurase output, consistent with a **hypomorphic** rather than null mechanism, and the deficit propagates to multiple Fe–S-dependent enzymes in both mitochondrial and cytosolic compartments.

> **Evidence type:** Human clinical + structural/in vitro. **Key PMIDs:** 31918395, 35026043.

Finding 3 — The phenotypic spectrum ranges from fatal infantile multisystem disease to milder adult-onset fatigue/GI disease

COXPD52 phenotype is variable and correlates with genotype. At the **severe infantile end** (homozygous p.Arg72Gln), patients present with neonatal/infantile-onset lactic acidemia, hypotonia, weakness, combined respiratory-chain complex II+III deficiency, abnormal mitochondrial morphology, and multisystem organ failure [PMID: 24498631]. Yang et al. (2022) reported that "Two infants with the same homozygous variant presented with hypotonia, weakness and lactic acidosis," with abnormal mitochondrial morphology on muscle biopsy and

decreased complex II (variably I/III) activity. At the **mild end**, the same study noted that "one patient with compound heterozygous p.(Arg72Gln) and p.(Arg412His) variants presented as a young adult with gastrointestinal symptoms and fatigue" [PMID: 35026043].

Suggested HPO terms: HP:0001942 (Metabolic acidosis) / HP:0003128 (Lactic acidosis); HP:0001252 (Hypotonia); HP:0003198 (Myopathy) / HP:0001324 (Muscle weakness); HP:0003200 (Ragged-red muscle fibers / abnormal mitochondria); HP:0011968 (Feeding difficulties); HP:0002024 (Malabsorption); HP:0012378 (Fatigue); HP:0003542 (Increased muscle lactate).

> **Evidence type:** Human clinical. **Key PMIDs:** 24498631, 35026043.

Finding 4 — NFS1 supplies sulfur for mitochondrial AND cytosolic Fe–S enzymes, tRNA thiolation, and genome maintenance; broad loss explains multisystem disease

NFS1 is the **sole mitochondrial cysteine desulfurase** and the rate-limiting enzyme of Fe–S cluster biogenesis. Its sulfur output feeds three major branches: (i) **mitochondrial Fe–S enzymes**—respiratory complexes I/II/III, mitochondrial aconitase (ACO2), and lipoic acid synthase; (ii) the **cytosolic iron–sulfur assembly (CIA) pathway**, which matures cytosolic and nuclear Fe–S proteins including cytosolic aconitase/IRP1 (ACO1) and multiple DNA replication/repair enzymes; and (iii) **tRNA thiolation**. Zhu et al. (2025) note that NFS1 "participates in numerous critical cellular processes, including iron-sulfur (Fe-S) cluster biosynthesis and tRNA thiolation" [PMID: 40141425].

The breadth of this dependency directly explains the multisystem clinical picture. Zangari et al. (2025) demonstrate that "NFS1 inhibition by D-Cys affects all cellular iron-sulfur cluster-dependent functions, including mitochondrial respiration, nucleotide metabolism and maintenance of genome integrity, leading to decreased oxygen consumption, DNA damage and cell cycle arrest" [PMID: 40797101]. Patient biochemistry confirms the combined mitochondrial + cytosolic aconitase deficiency predicted by this model [PMID: 35026043].

Suggested GO terms: GO:0016226 (iron-sulfur cluster assembly); GO:0031071 (cysteine desulfurase activity); GO:0097428 (protein maturation by iron-sulfur cluster transfer); GO:0002143 (tRNA wobble position thiouridine biosynthesis). **Cellular component:** GO:0005759 (mitochondrial matrix), GO:0005829 (cytosol). **CHEBI:** L-cysteine (CHEBI:17561), iron-sulfur cluster (CHEBI:30408).

> **Evidence type:** In vitro / cell biology + human clinical. **Key PMIDs:** 40797101, 40141425, 35026043.

Finding 5 — NFS1 is a PLP (vitamin B6)-dependent, intrinsically low-activity desulfurase requiring frataxin activation

NFS1 is a **pyridoxal 5′-phosphate (PLP, vitamin B6)-dependent** enzyme. Patra & Barondeau (2019) describe the core assembly machine as "a low-activity pyridoxal 5'-phosphate (PLP)-dependent cysteine desulfurase enzyme that consists of catalytic (NFS1), LYRM protein (ISD11), and acyl carrier protein (ACP) subunits" [PMID: 31511419]. The intrinsically low basal activity means the enzyme is critically dependent on accessory subunits and allosteric activators for physiological output.

Frataxin (FXN) is the key allosteric activator. Cory et al. (2025) describe how "Iron-sulfur clusters are essential protein cofactors synthesized in human mitochondria by an NFS1-ISD11-ACP-ISCU2-FXN assembly complex," with frataxin acting as a "molecular lock" that shifts the complex toward an active quaternary structure, accelerating substrate binding, persulfide formation, and sulfur delivery to ISCU2 [PMID: 39909887]. The functional interdependence of NFS1 with FXN is mechanistically important: it links COXPD52 biology to Friedreich ataxia (a frataxin-deficiency disease) and underscores why partial loss of NFS1 catalytic efficiency has outsized downstream consequences.

Protein annotations: UniProt Q9Y697 (NFS1_HUMAN); Pfam Aminotran_5 / class-V aminotransferase fold; InterPro cysteine desulfurase.

> **Evidence type:** Structural / biochemical. **Key PMIDs:** 31511419, 39909887.

Finding 6 — Ultra-rare, autosomal recessive, complete penetrance; p.Arg72Gln recurs via consanguinity/shared haplotype

Inheritance is **autosomal recessive**: two defective alleles are required, and heterozygous carriers are unaffected because a single functional allele provides sufficient desulfurase activity. The disease is **ultra-rare**, with only a handful of families reported since 2014—one consanguineous family identified through autozygosity mapping [PMID: 24498631] and three additional unrelated patients [PMID: 35026043]. The recurrent allele c.215G>A (p.Arg72Gln) appears in the homozygous state in consanguineous/isolated-population backgrounds ("we

identified c.215G>A, p.Arg72Gln in NFS1 as the likely causative mutation" via "autozygosity mapping, exome sequencing...") [PMID: 24498631], and as a compound heterozygous allele paired with c.1235G>A (p.Arg412His): "one patient with compound heterozygous p.(Arg72Gln) and p.(Arg412His) variants" [PMID: 35026043].

Notably, biallelic truncating/complete loss-of-function alleles have **not** been reported, consistent with *NFS1* being an essential gene—hypomorphic missense variants are tolerated (in the homozygous or compound-heterozygous state), whereas a complete null is predicted to be embryonic-lethal. ClinVar lists *NFS1* c.215G>A as pathogenic/likely pathogenic for this condition. No formal Orphanet prevalence, incidence, founder-population frequency, or carrier-frequency estimate has been published.

> **Evidence type:** Human genetic / population. **Key PMIDs:** 24498631, 35026043.

Finding 7 — NFS1/cysteine desulfurase is deeply conserved; orthologs are essential across species

Fe–S clusters and the cysteine desulfurase that builds them are, in the words of Farhan et al. (2014), "a class of highly conserved and ubiquitous prosthetic groups" [PMID: 24498631]. NFS1 orthologs span the tree of life: *S. cerevisiae* NFS1 (essential for viability; mitochondrial), *E. coli* IscS/NifS (bacterial cysteine desulfurases that are the structural and functional prototypes), *Drosophila melanogaster* Nfs1, and mouse *Nfs1* (NCBI Gene 18041). The machinery's functional interdependence with conserved partners frataxin (FXN/Yfh1), ISCU (Isu1/2), ISD11/LYRM4, and HSPA9/Mortalin has been demonstrated in yeast and zebrafish ISC-biogenesis models. Shan & Cortopassi (2016) showed that "HSPA9 interacts with and stabilizes the mitochondrial ISC biogenesis proteins frataxin, Nfs1, ISCU, and Nfu," linking Nfs1 to aconitase activity and erythroid differentiation [PMID: 26702583].

NCBI Taxa: human (9606), mouse (10090), zebrafish (7955), *Drosophila* (7227), *S. cerevisiae* (4932), *E. coli* (562). The essentiality of NFS1 orthologs means complete-knockout model organisms are typically inviable; disease-relevant models require hypomorphic, conditional, or knock-in (patient-variant) approaches.

> **Evidence type:** Comparative / model organism. **Key PMIDs:** 24498631, 26702583.

Finding 8 — Diagnosis rests on exome/genome sequencing plus biochemical Fe–S enzyme deficiency; treatment is supportive with poor infantile prognosis

Diagnosis is established by whole-exome or whole-genome sequencing confirming biallelic *NFS1* variants (NFS1 is not included on all targeted mitochondrial gene panels), supported by a distinctive biochemical signature: elevated blood/CSF lactate (and lactate:pyruvate ratio), muscle biopsy showing abnormal mitochondrial morphology, and respiratory-chain enzyme assays revealing combined complex II+III (±I) deficiency **plus** reduced activity of both mitochondrial and cytosolic aconitase. Yang et al. (2022) summarize the signature: "Skeletal muscle biopsy from patients 1 and 3 showed abnormal mitochondrial morphology, and functional analyses demonstrated decreased activity in respiratory chain complex II and variably in complexes I and III. We found decreased mitochondrial and cytosolic aconitase activities" [PMID: 35026043]. Because the defect spans complexes II/III and *extra*-mitochondrial Fe–S enzymes, the biochemical pattern distinguishes COXPD52 from mtDNA-encoded complex defects, which do not impair cytosolic aconitase.

Differential diagnosis includes other Fe–S biogenesis disorders: ISCU myopathy, Friedreich ataxia (FXN), LYRM4/ISD11 deficiency (COXPD19), the multiple-mitochondrial-dysfunction syndromes (BOLA3, NFU1, IBA57, ISCA2), and FDX2/FDXR disorders.

Treatment: there is no disease-specific or approved therapy. Management is supportive mitochondrial care—treating lactic acidosis, providing nutritional support, administering a "mitochondrial cocktail" of vitamins/cofactors, and avoiding metabolic decompensation. **Prognosis:** severe infantile homozygous p.Arg72Gln cases have high early mortality from multisystem organ failure [PMID: 24498631], whereas the milder compound-heterozygous genotype survived to adulthood [PMID: 35026043].

Suggested NCIT concepts: Supportive Care (NCIT:C15277), Genetic Counseling (NCIT:C15417).

> **Evidence type:** Human clinical. **Key PMIDs:** 35026043, 24498631.

Section-by-Section Disease Characteristics

1. Disease Information

COXPD52 is a **nuclear-gene mitochondrial disease** of iron–sulfur cluster biogenesis. It was originally described (2014) as "infantile mitochondrial complex II/III deficiency" and is now cataloged in the OMIM "combined oxidative phosphorylation deficiency" series as entry **#619386**.

Identifier	Value
OMIM	#619386
MONDO	MONDO:0030311
Gene	NFS1 (HGNC:7808; NCBI Gene 9054; UniProt Q9Y697)
Locus	20q11.22
Synonym	COXPD52; infantile mitochondrial complex II/III deficiency
Orphanet / ICD-10 / ICD-11 / MeSH	No dedicated code identified (subsumed under mitochondrial/ OXPHOS disorders)

Information source: derived almost entirely from **individual-patient case reports** (two primary publications describing a total of four patients) rather than aggregated disease-level registries—reflecting the disorder's ultra-rarity.

2. Etiology

Causal factor: purely **genetic**—biallelic hypomorphic variants in *NFS1*. **Genetic risk factors:** the recurrent c.215G>A (p.Arg72Gln) allele and the c.1235G>A (p.Arg412His) allele; consanguinity is a major contributor by increasing homozygosity for the recurrent variant [PMID: 24498631]. No environmental, infectious, lifestyle, or occupational risk factors are established. No protective variants or modifier alleles are documented. **Gene–environment interactions:** none established; however, intercurrent catabolic stress (infection, fasting) may precipitate metabolic decompensation in a vulnerable Fe–S-deficient background (mechanistically inferred, not formally demonstrated).

3. Phenotypes

Phenotype	Type	HPO	Onset	Severity	Frequency
Lactic acidosis	Lab abnormality	HP:0003128	Neonatal/ infantile	Severe (infantile)	Core feature
Hypotonia	Clinical sign	HP:0001252	Infantile	Moderate–severe	Core feature
Muscle weakness / myopathy	Clinical sign	HP:0001324 / HP:0003198	Infantile	Variable	Common
Abnormal mitochondrial morphology	Pathology/ biopsy	HP:0003200 (ragged-red)	Infantile	—	Reported in biopsied patients
Multisystem organ failure	Clinical	—	Infantile	Severe/ fatal	Severe genotype
Gastrointestinal symptoms	Symptom	HP:0011024	Young adult	Mild	Mild genotype
Fatigue	Symptom	HP:0012378	Young adult	Mild	Mild genotype
Increased muscle lactate	Lab abnormality	HP:0003542	Infantile	—	Reported

Quality-of-life impact: severe infantile disease is devastating with high early mortality; the milder adult phenotype (chronic GI symptoms, fatigue) impairs daily functioning but is compatible with survival. Formal EQ-5D/SF-36 data are unavailable given rarity.

4. Genetic / Molecular Information

Causal gene: *NFS1* (disease OMIM #619386). **Pathogenic variants** reported: c.215G>A (p.Arg72Gln)—recurrent, missense, pathogenic/likely pathogenic in ClinVar; and c.1235G>A (p.Arg412His)—missense, in trans with p.Arg72Gln in the adult patient. **Variant class:** missense (hypomorphic). **Functional consequence:** partial **loss of function** (reduced cysteine desulfurase activity); no gain-of-function or dominant-negative mechanism. **Somatic vs germline:** germline. **Allele frequency:** p.Arg72Gln is rare in gnomAD, consistent with a pathogenic recessive allele. **Modifier genes:** none established, though genes of the ISC/CIA

machinery (FXN, ISCU, LYRM4, FDX2, HSPA9) are plausible candidates. **Epigenetic changes / chromosomal abnormalities:** none reported or expected for this single-gene point-mutation disorder.

5. Environmental Information

Not applicable. No environmental toxins, radiation, pollution, occupational exposures, lifestyle factors, or infectious agents are implicated in causing COXPD52. The disease is monogenic and recessive.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic hypomorphic *NFS1* variants** (e.g., p.Arg72Gln) **result in** a structurally intact but catalytically impaired cysteine desulfurase within the core ISC complex (NFS1–ISD11–ACP–ISCU2, activated by FXN; electrons from FDX2) [PMID: 31511419, 39909887].
2. Reduced NFS1 activity **leads to** diminished mobilization of sulfur from L-cysteine and **decreased de novo assembly of [2Fe–2S] clusters** on the ISCU2 scaffold [PMID: 31918395].
3. The resulting Fe–S cofactor shortfall **branches** into multiple downstream failures:
4. **Branch A (mitochondrial respiration):** failure to mature Fe–S subunits of respiratory complexes I, II, and III **results in** combined OXPHOS deficiency and impaired electron transport [PMID: 24498631, 35026043].
5. **Branch B (TCA cycle / metabolism):** loss of Fe–S in mitochondrial aconitase (ACO2) and reduced lipoic acid synthesis **impair** the TCA cycle and oxidative decarboxylation (lipoylation only mildly affected in patients) [PMID: 35026043].
6. **Branch C (cytosolic/nuclear):** reduced sulfur supply to the CIA pathway **leads to** decreased cytosolic aconitase (ACO1/IRP1) activity and impaired maturation of DNA replication/repair Fe–S enzymes, causing DNA damage and cell-cycle arrest (demonstrated for NFS1 inhibition) [PMID: 40797101].
7. **Branch D (translation):** impaired tRNA thiolation **compromises** accurate mitochondrial/cytosolic translation (inferred for patients; demonstrated as an NFS1 function) [PMID: 40141425].
8. Impaired OXPHOS and TCA flux **cause** a shift to anaerobic glycolysis, **producing lactic acidemia** and energy deficit [PMID: 24498631].

9. Energy failure in high-demand tissues (skeletal/cardiac muscle, CNS, liver) **results in** hypotonia, weakness, abnormal mitochondrial morphology, and—at the severe end—**multisystem organ failure** [PMID: 24498631, 35026043].

Upstream vs downstream: the *NFS1* catalytic defect is the single upstream lesion; all respiratory, metabolic, genome-maintenance, and translational phenotypes are downstream consequences of a shared Fe–S supply bottleneck. **Cell types / processes:** high-oxidative-demand cells (skeletal myocytes CL:0000188, cardiomyocytes CL:0000746, hepatocytes CL:0000182, neurons CL:0000540). **GO processes:** GO:0016226 (iron-sulfur cluster assembly), GO:0006099 (TCA cycle), GO:0045333 (cellular respiration), GO:0006281 (DNA repair). **Subcellular compartments:** mitochondrial matrix (GO:0005759), cytosol (GO:0005829).

7. Anatomical Structures Affected

Primary organs: skeletal muscle (UBERON:0001134), and—via systemic energy failure—heart, liver, and central nervous system. **Body systems:** musculoskeletal, nervous, metabolic/endocrine, gastrointestinal. **Tissue level:** striated muscle tissue; the adult phenotype implicates the GI tract (UBERON:0001555). **Cell level:** skeletal myocytes (CL:0000188), neurons (CL:0000540), hepatocytes (CL:0000182). **Subcellular:** mitochondria (GO:0005739) are the principal affected compartment, with secondary cytosolic/nuclear involvement via the CIA pathway. **Lateralization:** systemic/bilateral (not a lateralized disorder).

8. Temporal Development

Onset: bimodal and genotype-dependent—**neonatal/infantile** (severe homozygous p.Arg72Gln) vs **young adult** (compound heterozygous p.Arg72Gln/p.Arg412His) [PMID: 24498631, 35026043]. **Onset pattern:** acute/subacute metabolic crisis in infants; insidious chronic course in the adult. **Progression:** rapidly progressive to multisystem organ failure in the infantile form; chronic and comparatively stable in the mild form. **Disease course:** progressive (infantile) vs chronic (adult). **Critical period:** the neonatal/early-infantile window is the period of greatest vulnerability and the key opportunity for supportive intervention.

9. Inheritance and Population

Inheritance: autosomal recessive, **complete penetrance** in biallelic individuals, with **variable expressivity** driven by genotype (homozygous vs compound-heterozygous). **Epidemiology:** ultra-rare—only four patients in two publications; no published prevalence or incidence. **Carrier frequency:** not formally estimated; p.Arg72Gln is rare in gnomAD. **Consanguinity:** a documented contributor (founding family identified by autozygosity

mapping) [PMID: 24498631]. **Founder effect:** the recurrence of p.Arg72Gln across unrelated families suggests a possible shared ancestral haplotype, though this has not been formally proven. **Anticipation / germline mosaicism:** not applicable (non-repeat, point-mutation disorder). **Sex ratio:** no sex bias reported (autosomal). Note that biallelic null genotypes are absent, consistent with *NFS1* essentiality.

10. Diagnostics

Genetic testing (definitive): WES or WGS to identify biallelic *NFS1* variants; single-gene/targeted testing once *NFS1* is suspected. NFS1 is not on all mitochondrial panels, so broad sequencing is preferred. **Biochemical/laboratory:** elevated blood and CSF lactate with increased lactate:pyruvate ratio; muscle respiratory-chain enzyme assays showing combined complex II+III (±I) deficiency; and the discriminating finding of **reduced mitochondrial *and* cytosolic aconitase activity** [PMID: 35026043]. **Histopathology:** muscle biopsy showing abnormal mitochondrial morphology [PMID: 35026043, 24498631]. **Differential diagnosis:** ISCU myopathy, Friedreich ataxia (FXN), COXPD19 (LYRM4/ISD11), multiple mitochondrial dysfunction syndromes (NFU1, BOLA3, IBA57, ISCA2), FDX2/FDXR disorders—distinguished by gene and by the characteristic combined mito+cytosolic Fe–S enzyme pattern. **Screening:** no newborn or population carrier-screening program exists; cascade testing of relatives and prenatal/preimplantation testing are options once a familial genotype is known.

11. Outcome / Prognosis

Survival/mortality: the severe infantile homozygous form carries **high early mortality** from multisystem organ failure [PMID: 24498631]. The milder compound-heterozygous genotype is compatible with **survival to adulthood** [PMID: 35026043]. **Morbidity:** infantile survivors face profound disability from energy-failure myopathy/encephalopathy; the adult form causes chronic GI symptoms and fatigue. **Prognostic factors:** genotype is the dominant prognostic determinant (homozygous p.Arg72Gln = severe; compound heterozygous with p.Arg412His = mild), alongside age of onset and the severity of lactic acidosis. No validated prognostic biomarkers beyond lactate and enzyme-activity measures are established.

12. Treatment

No disease-specific or approved therapy exists. Management is **supportive mitochondrial care** (NCIT:C15277, Supportive Care):

- **Pharmacotherapy/supportive:** correction of lactic acidosis; "mitochondrial cocktail" cofactors/vitamins (e.g., riboflavin, thiamine, coenzyme Q10, L-carnitine)—empiric, not disease-validated.
- **Pharmacogenomics / gene therapy / cell therapy / RNA therapy / targeted or immunotherapy:** none established or in trials for COXPD52.
- **Nutritional/metabolic:** avoidance of catabolic stress and fasting; nutritional support for feeding difficulties.
- **Genetic counseling** (NCIT:C15417) for families.

Mechanistic note: Because NFS1 is a **PLP (vitamin B6)-dependent** enzyme [PMID: 31511419] and depends on cysteine as substrate, cofactor/substrate-oriented strategies (e.g., vitamin B6, cysteine/N-acetylcysteine supplementation) are biologically rational hypotheses worth formal study—analogous to the reported benefit of L-cysteine/NAC in the related Fe–S/tRNA disorder COXPD10 [PMID: 42753170]—but they remain **unproven for COXPD52**.

13. Prevention

Primary prevention is limited to reproductive genetic strategies: **genetic counseling**, carrier/cascade testing in affected families, and **prenatal or preimplantation genetic diagnosis** once the familial variants are known. **Secondary/tertiary prevention** centers on early recognition of metabolic decompensation and proactive supportive management to prevent complications. No vaccine, environmental, or public-health intervention is applicable (non-infectious, non-environmental monogenic disease).

14. Other Species / Natural Disease

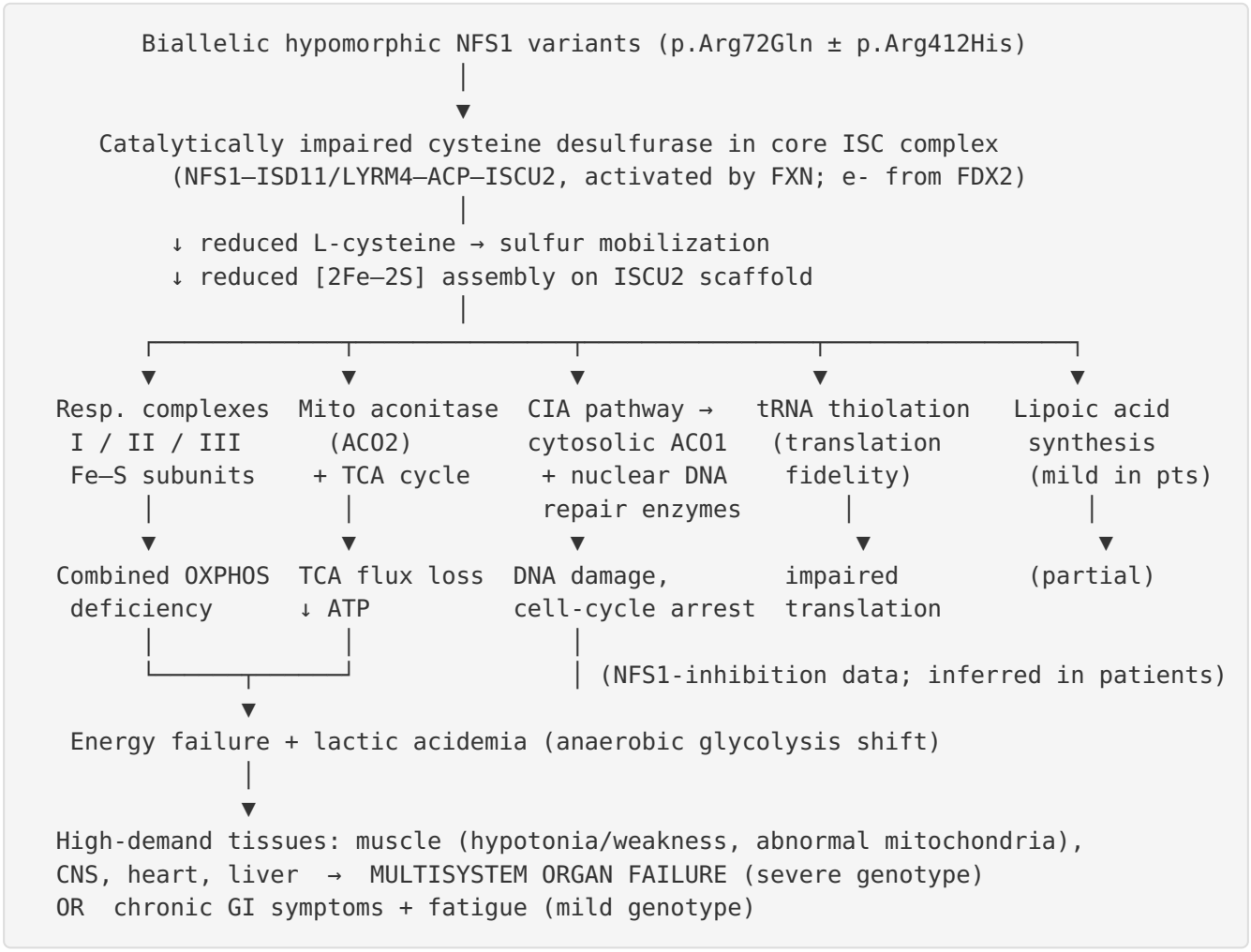
NFS1/cysteine desulfurase is **deeply conserved and essential** across eukaryotes and bacteria [PMID: 24498631]. Orthologs: *S. cerevisiae* NFS1, *E. coli* IscS/NifS, *Drosophila* Nfs1, mouse *Nfs1* (NCBI Gene 18041; Taxon 10090), zebrafish (Taxon 7955). **NCBI Taxa:** human 9606, mouse 10090, zebrafish 7955, *Drosophila* 7227, *S. cerevisiae* 4932, *E. coli* 562. No naturally occurring companion-animal or wildlife disease equivalent to COXPD52 is cataloged (e.g., in OMIA). There is no zoonotic or cross-species transmission dimension. Evolutionary conservation of the ISC machinery (NFS1 with FXN/Yfh1, ISCU/Isu1-2, ISD11/LYRM4, HSPA9) makes model organisms highly informative for mechanism [PMID: 26702583].

15. Model Organisms

Model types: because NFS1 orthologs are essential, complete knockouts are inviable; disease-relevant modeling uses **hypomorphic, conditional, or patient-variant knock-in** approaches, plus **cellular/in vitro** systems. Established systems informing COXPD52 biology include: - **Yeast (*S. cerevisiae*):** NFS1/Yfh1 ISC-biogenesis genetics; frataxin-bypass and complementation studies [PMID: 40563426]. - **Zebrafish / erythroid models:** Nfs1 within the HSPA9-stabilized ISC machinery; required for aconitase activity and erythroid differentiation [PMID: 26702583]. - **Mammalian cell lines:** NFS1 knockdown/inhibition (e.g., D-cysteine) recapitulates loss of Fe–S-dependent respiration, nucleotide metabolism, and genome integrity [PMID: 40797101]; in vitro reconstitution of the NFS1–ISD11–ACP–ISCU2–FXN complex enables direct enzymatic study of patient variants [PMID: 31511419, 39909887].

Phenotype recapitulation: cellular and lower-organism models faithfully reproduce the **biochemical** core (loss of Fe–S enzyme activities, respiratory deficiency, aconitase loss), supporting mechanism. **Limitations:** no published rodent model carries the patient p.Arg72Gln allele to recapitulate the full infantile multisystem syndrome; the organismal phenotype and genotype–phenotype gradient remain incompletely modeled. **Resources:** MGI (mouse *Nfs1*), SGD (yeast NFS1), ZFIN (zebrafish), Cellosaurus (cell lines).

Mechanistic Model / Interpretation



The unifying interpretation is that COXPD52 is a **single upstream catalytic bottleneck disease**: one partially-active enzyme (NFS1) sits at the head of the entire cellular Fe–S supply chain, so a modest reduction in its activity produces a *combined, multi-pathway, multi-organ* deficiency. The biochemical fingerprint that distinguishes it from mtDNA-encoded respiratory diseases—**simultaneous mitochondrial and cytosolic aconitase loss**—is the direct readout of this upstream-bottleneck architecture. Genotype tunes severity: the more residual desulfurase activity the alleles preserve, the milder and later the disease.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Supports
24498631	*Exome sequencing identifies NFS1 deficiency in infantile complex II/III deficiency*	Human genetic/ clinical	F1, F2 (disease def.), F3 (severe phenotype), F6 (AR, consanguinity), F7 (conservation), F8 (Dx/prognosis)
35026043	*Expanding the phenotypic and molecular spectrum of NFS1-related disorders*	Human clinical	F1 (recurrence), F2 (biochemistry), F3 (spectrum), F4 (mito+cyto aconitase), F6 (compound het), F8 (Dx signature)
31918395	*Mammalian Fe-S cluster biogenesis: frataxin, ACP, ATPase transfer*	Review/ structural	F2 (core complex, cluster assembly on ISCU)
40797101	*D-cysteine impairs tumour growth by inhibiting NFS1*	In vitro	F4 (breadth: respiration, nucleotide metabolism, genome integrity)
40141425	*Selective inhibitor of human NFS1 via virtual screening*	In vitro	F4 (Fe-S biosynthesis + tRNA thiolation)
31511419	*Mechanism of activation of human cysteine desulfurase by frataxin*	Biochemical	F5 (PLP-dependent, low-activity; subunits)
39909887	*Frataxin traps low-abundance quaternary structure*	Structural	F5 (NFS1-ISD11-ACP-ISCU2-FXN complex)
26702583	*Mitochondrial Hspa9/ Mortalin regulates erythroid differentiation via ISC assembly*	Model organism	F7 (conserved ISC machinery incl. Nfs1)
23814038	*Mutations in LYRM4/ISD11 cause combined OXPHOS deficiency*	Human clinical	Differential Dx (COXPD19); NFS1 stabilization context
42753170	*L-Cysteine/NAC improves outcome in COXPD10*	Human case	Rational (unproven) cofactor-therapy analogy

Challenges / cautionary context: The large body of **NFS1 cancer literature** (osteosarcoma, colorectal, lung, AML, DLBCL, OSCC—PMIDs 42704489, 42069097, 41005206, 42878337, 35221331, 42068882, 42034254) studies NFS1 as an *overexpressed, pro-survival, anti-ferroptosis* oncogenic dependency. This is biologically consistent with (but opposite in direction to) COXPD52: in cancer, NFS1 is *up*regulated to protect against ferroptosis and support proliferation, whereas in COXPD52 NFS1 is *hypo*functional. These papers reinforce the enzyme's central Fe–S role but do not directly describe the recessive disease.

Limitations and Knowledge Gaps

1. **Very small evidence base.** The entire clinical literature on COXPD52 comprises two publications and four patients. Genotype–phenotype correlation, penetrance nuances, and natural history are therefore based on minimal data.
2. **No epidemiology.** No published prevalence, incidence, carrier frequency, founder-haplotype confirmation, or geographic distribution exists.
3. **Narrow variant spectrum.** Only two missense variants (p.Arg72Gln, p.Arg412His) are reported; the full allelic and phenotypic spectrum (including whether any hypomorphic truncating combinations are viable) is unknown. Absence of biallelic null genotypes is inferred to reflect essentiality but is not formally proven in humans.
4. **No disease-specific therapy or trials.** Cofactor/substrate strategies (vitamin B6, cysteine/NAC) are mechanistically plausible but untested in COXPD52.
5. **No dedicated animal model of the human disease.** Existing models are essential-gene knockouts (inviable) or knockdowns; no published rodent carries the patient p.Arg72Gln allele to recapitulate the multisystem syndrome.
6. **Mechanistic steps partly inferred.** The contributions of tRNA-thiolation and genome-maintenance branches to patient phenotype are extrapolated from NFS1-inhibition/cancer studies rather than demonstrated in patient tissue.

Proposed Follow-up Experiments / Actions

1. **Build a patient-variant knock-in model.** Generate a *Nfs1* p.Arg72Gln (and compound-heterozygous) mouse or zebrafish, or patient iPSC-derived cardiomyocytes/myotubes, to recapitulate the organismal phenotype and enable natural-history and therapeutic studies.
2. **Quantitative enzymology of patient variants.** Reconstitute NFS1–ISD11–ACP–ISCU2–FXN with p.Arg72Gln and p.Arg412His to measure residual desulfurase activity, cluster-transfer rate, and FXN-responsiveness—directly testing the genotype–severity gradient.
3. **Test cofactor/substrate therapies.** In patient cells and models, evaluate high-dose vitamin B6 (PLP), L-cysteine/N-acetylcysteine, and iron-handling modulators, mirroring the COXPD10 cysteine/NAC precedent [PMID: 42753170].
4. **Establish a patient registry / GeneMatcher outreach.** Aggregate additional cases to define prevalence, allelic spectrum, penetrance, and genotype–phenotype correlation.
5. **Validate the diagnostic biochemical signature.** Prospectively test whether combined mitochondrial + cytosolic aconitase deficiency reliably discriminates COXPD52 (and other ISC/CIA disorders) from mtDNA-encoded OXPHOS defects in diagnostic cohorts.
6. **Confirm or refute a founder haplotype.** Haplotype-phase the recurrent c.215G>A allele across reported families to establish whether it reflects a shared ancestral origin or recurrent mutation.

Report compiled from 8 confirmed findings across 35 reviewed papers over 5 investigation iterations. Evidence types are labeled per claim (human clinical, model organism, in vitro/biochemical, structural, computational). Primary clinical evidence derives from two publications (P 24498631, P 35026043); mechanistic evidence from Fe–S biogenesis structural/biochemical and NFS1-function studies.
